PROBLEM CL17

Airplane with an Engine without Moving Parts: Exit State, Thrust, Torque and Canard

Original statement · CL17.pdf

Thrust | Angular momentum | Energy | Trim

OFFICIAL PROBLEM STATEMENT

An experimental airplane is powered by an engine without moving parts that delivers a heat $\dot{Q}$ per unit time. The entrance and exit sections of the engine are, respectively, below and above the airplane's center of gravity (CoG), which is placed at the origin of the coordinate system shown in the figure of CL17.pdf. Both sections are rectangular with the same spanwise length b (along the z-direction), and heights a_i and a_o, as indicated in the figure. The flow at the engine's inlet has velocity V_i, pressure p_i and density ρ_i, while the outlet jet discharges at pressure p_o. The flow is steady; gravitational forces are negligible; the conditions at the engine's inlet and outlet sections are uniform; the air behaves as a perfect gas of constant R_g and specific heat ratio γ; the known data are $(a_i, V_i, \rho_i, p_i, a_o, p_o, p_a, \dot{Q})$.

- Applying the mass and energy conservation equations to an appropriately defined control volume, obtain the pair of equations that determine the jet exhaust velocity, V_o, and the corresponding density, ρ_o. In the remainder of the problem, assume that V_o and ρ_o are already known.
- Use the momentum equation to obtain the thrust provided by the engine, $\bar{T}$.
- Use the angular momentum equation to obtain the net torque $\bar{M}$ acting on the airplane with respect to its CoG. Assume that the surrounding air is at uniform pressure p_a, except at the engine's inlet and outlet.
- To compensate the torque obtained in the previous question, the engineers decide that the main wing's CoG coincides with that of the full airplane, so that it doesn't contribute to the torque, and to place a control surface (canard) at a distance L from the CoG (see figure). Knowing that this control surface provides a lift force F, determine the distance L.

Source: CL17.pdf (page 1, exam of June 27, 2013; statement only, no official solution in the source). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady; neglect $\bar{g}$; uniform properties at the inlet and outlet; perfect gas, $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$; no moving parts (no shaft work); the heat $\dot{Q}$ is the only energy input.
- Surrounding air at p_a everywhere except at the two sections; viscous stresses negligible on the sections.
- Geometry (reading of the figure): the inlet section spans $-a_i < y < 0$ and the outlet section $0 < y < a_o$, both normal to $\bar{e}_x$, with the flow along $+\bar{e}_x$ (from the nose side to the tail). The nose points towards $-\bar{e}_x$; the canard is at $x = -L$.
- Data: $[a_i,\ V_i,\ \rho_i,\ p_i,\ a_o,\ p_o,\ p_a,\ \dot{Q},\ b,\ \gamma,\ F]$. Unknowns: $[V_o,\ \rho_o,\ \bar{T},\ \bar{M},\ L]$.

Symbol	Meaning	SI unit
a_i, a_o, b	section heights and span	m
V_i, ρ_i, p_i	inlet state	m/s, kg/m^3, Pa
V_o, ρ_o, p_o	outlet state	m/s, kg/m^3, Pa
$\dot{Q}$	heat added per unit time	W
$\bar{T}, \bar{M}$	thrust and torque on the airplane	N, N·m

No official solution

CL17.pdf contains only the statement. The solution below is derived independently and checked numerically; there is no official key to compare with.

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the airplane (steady flight or test bench), origin at the CoG, $\bar{e}_x$ towards the tail, $\bar{e}_y$ upwards, $\bar{e}_z$ towards the reader. Fixed control volume: the air inside the engine.
- Surfaces: Σ_i (inlet, $\bar{n} = -\bar{e}_x$, area $a_i b$), Σ_o (outlet, $\bar{n} = +\bar{e}_x$, area $a_o b$), Σ_w (engine walls, at rest).

$$\boxed{\bar{T} = \int_{\Sigma_w} \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\bar{M} = \int_{\Sigma_w} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

1. Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\, dV}}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \left(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0} \right) \cdot \bar{n}\, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\, \bar{v} \cdot \bar{n}\, d\sigma$
Σ_i	$-\rho_i V_i a_i b$
Σ_o	$+\rho_o V_o a_o b$
Σ_w	0

$$\boxed{\rho_i V_i a_i = \rho_o V_o a_o}, \qquad \dot{m} = \rho_i V_i a_i b \tag{1}$$

1. Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\left(e+\frac{v^2}{2}\right) dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e+\frac{v^2}{2}\right)\left[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\cancel{\int_{V_c} \rho\bar{g}\cdot\bar{v}\,dV}}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\,d\sigma + \int_{V_c} Q_R\,dV$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_i	$-\dot{m}\left(e_i+\frac{V_i^2}{2}\right)$	$+p_i V_i a_i b$	0	0
Σ_o	$+\dot{m}\left(e_o+\frac{V_o^2}{2}\right)$	$-p_o V_o a_o b$	0	0
Σ_w	0	0 ($\bar{v}=0$)	0	$-\int_{\Sigma_w}\bar{q}\cdot\bar{n}\,d\sigma$

Boxed term plus the volume heat release: $\dot{Q}$.

$$\boxed{\dot{m}\left[\left(\frac{\gamma}{\gamma-1}\frac{p_o}{\rho_o}+\frac{V_o^2}{2}\right)-\left(\frac{\gamma}{\gamma-1}\frac{p_i}{\rho_i}+\frac{V_i^2}{2}\right)\right]=\dot{Q}} \qquad (2)$$

(1) and (2) are the pair of equations for V_o and ρ_o. Explicitly (extension): with (1), $\frac{p_o}{\rho_o} = \frac{p_o V_o a_o b}{\dot{m}}$, and (2) becomes $\frac{V_o^2}{2} + BV_o - H = 0$ with $B = \frac{\gamma}{\gamma-1}\frac{p_o a_o b}{\dot{m}}$ and $H = \frac{\gamma}{\gamma-1}\frac{p_i}{\rho_i} + \frac{V_i^2}{2} + \frac{\dot{Q}}{\dot{m}}$, whose positive root is $V_o = -B + \sqrt{B^2+2H}$.

2. Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\left[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right]d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect } \bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_i	$-\dot{m}V_i\,\bar{e}_x$	$+(p_i-p_a)a_i b\,\bar{e}_x$	0
Σ_o	$+\dot{m}V_o\,\bar{e}_x$	$-(p_o-p_a)a_o b\,\bar{e}_x$	0
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{T}$.

$$\boxed{\bar{T} = -\left[\dot{m}\left(V_o - V_i\right) + (p_o - p_a)a_o b - (p_i - p_a)a_i b\right]\bar{e}_x} \tag{3}$$

3. Angular momentum about the CoG, z component (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\,\bar{x}\wedge\bar{v}\,dV}}_{0} + \int_{\Sigma_c} \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c} \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})_z(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int\left[\bar{x}\wedge(p-p_a)\bar{n}\right]_z d\sigma$	viscous
Σ_i	$\int_{-a_i}^{0}\rho_i(-yV_i)(-V_i)\,b\,dy = -\rho_i V_i^2\frac{a_i^2}{2}b$	$-\int_{-a_i}^{0}(p_i-p_a)\,y\,b\,dy = +(p_i-p_a)\frac{a_i^2}{2}b$	0
Σ_o	$\int_0^{a_o}\rho_o(-yV_o)(V_o)\,b\,dy = -\rho_o V_o^2\frac{a_o^2}{2}b$	$-\int_0^{a_o}(p_o-p_a)(-y)\,b\,dy = +(p_o-p_a)\frac{a_o^2}{2}b$	0
Σ_w	0	$-\int_{\Sigma_w}\left[\bar{x}\wedge(p-p_a)\bar{n}\right]_z d\sigma$	$\int_{\Sigma_w}\left[\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\right]_z d\sigma$

Boxed terms: $-M_z$. On a section normal to $\bar{e}_x$, $(\bar{x}\wedge\bar{v})_z = xv_y - yv_x = -yV$ and $(\bar{x}\wedge\bar{n})_z = -yn_x$. Antiderivative of y: $y^2/2$; Barrow's rule gives $\left[\frac{y^2}{2}\right]_{-a_i}^{0} = -\frac{a_i^2}{2}$ and $\left[\frac{y^2}{2}\right]_{0}^{a_o} = \frac{a_o^2}{2}$.

$$\boxed{\bar{M} = \frac{b}{2}\left[\left(\rho_i V_i^2 + p_i - p_a\right)a_i^2 + \left(\rho_o V_o^2 + p_o - p_a\right)a_o^2\right]\bar{e}_z} \tag{4}$$

4. Canard position

The lift $F\bar{e}_y$ of the canard, applied at $x = -L$, has moment $(-L\bar{e}_x) \wedge (F\bar{e}_y) = -LF\,\bar{e}_z$ about the CoG. The wing gives no moment (its CoG is the airplane's). Trim requires $\bar{M} - LF\bar{e}_z = 0$:

$$\boxed{L = \frac{M_z}{F} = \frac{b}{2F}\left[\left(\rho_i V_i^2 + p_i - p_a\right) a_i^2 + \left(\rho_o V_o^2 + p_o - p_a\right) a_o^2\right]} \tag{5}$$

PHASE 4

Interpretation, Limits and Dimensional Check

- The airflow enters below the CoG and leaves above it, so both the intake (pushed backwards) and the exhaust reaction (pushed forwards) create a positive M_z, which pitches the nose down (the nose is at $-x$). The canard ahead of the CoG with an upward lift balances it, which is why $L > 0$.
- Heating raises the stagnation enthalpy (2): the gas leaves faster ($V_o > V_i$), giving the thrust (3) along $-\bar{e}_x$ (forwards).
- Dimensions (SI): $[\dot{m}V] = \mathrm{N}$; $[(\rho V^2 + p)a^2 b] = \mathrm{N \cdot m}$; $[M_z/F] = \mathrm{m}$.
- Numerical illustration (hypothetical data): $a_i = 0.3$ m, $a_o = 0.25$ m, $b = 1$ m, $V_i = 150$ m/s, $\rho_i = 1$ kg/m^3, $p_i = 0.9 \times 10^5$ Pa, $p_o = p_a = 10^5$ Pa, $\dot{Q} = 20$ MW, $\gamma = 1.4$, $F = 5$ kN give $V_o = 362.56$ m/s, $\rho_o = 0.49647$ kg/m^3, $|\bar{T}| = 12565$ N, $M_z = 2601.9$ N·m, $L = 0.5204$ m (energy residual below 10^{-7}).